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Arnaud Blouin & Benoit Combemale (Univ Rennes, IRISA, Inria)
Nadine Couture (ESTIA, LaBRI)

Subject: Application for PhD Position – Exploring alternative evolutions of digital twins (PC5)

Dear Arnaud Blouin, Benoit Combemale, and Nadine Couture,

I am applying for the PC5 PhD position on "Exploring alternative evolutions of digital twins" hosted by the DiverSE research group and ESTIA. With my strong background in Artificial Intelligence, autonomous system behavior, and advanced 3D web visualizations, I am highly motivated to tackle the challenge of effectively representing and exploring the branching, alternative timelines of complex digital twins.

During the engineering of my "Autonomous Cyber-Shield Platform" and the generative Mamba-SSM simulation framework, I encountered the exact problem of exploring "What-If" scenarios. Predicting future system states often results in multiple probable outcomes that diverge significantly based on minute parameter shifts. To manage this, I utilized Adversarial Co-Evolution to simulate thousands of branching paths. However, the visualization and interactive exploration of these diverging timelines remain a significant HCI challenge. I am eager to apply model-driven engineering principles to decouple these evolutionary datasets from their visual representation, allowing us to build an interactive dashboard that intuitively visualizes these alternative paths without cognitive overload.

My hands-on experience building high-performance Next.js and Three.js visual interfaces positions me perfectly to translate the complex multi-dimensional data of digital twin divergence into an accessible, interactive format. I am excited to collaborate with the DiverSE and ESTIA teams to integrate these exploratory frameworks into Artemis and validate them against real-world industrial platforms. Please find attached my resume and transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui